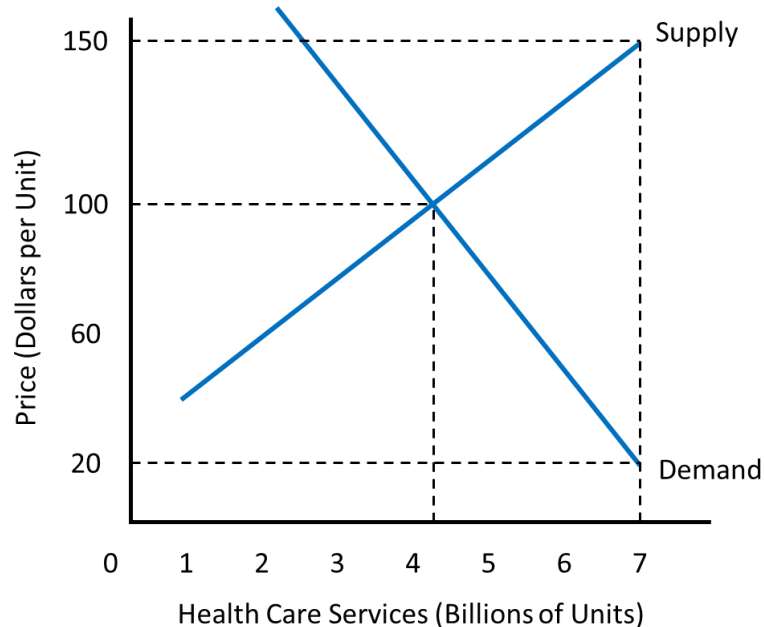


財政學 HW7

1. (Hyman 9.2) The graph on the bottom shows the supply and demand for health care in the nation. The demand curve reflects the marginal social benefit of health care, while the supply curve reflects the marginal social cost.

What is the efficient amount of health care expenditures?

Suppose that under current arrangements health insurance reduces the price of health care to the population from an average of \$100 per unit to \$20 per unit. Show how third-party payments will affect the market for health care by affecting the quantity demanded and quantity supplied. Assuming that the price must increase \$150 to accommodate the new quantity demanded, calculate the increase in expenditures for health care. Is efficiency now attained in market?



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a. \$400 billion

b. The quantity demanded and supplied will increase to 7 billion units.

Expenditures increase to \$1,050 trillion—an increase of \$650 billion. Efficiency is no longer attained because MSC exceeds MSB at the new output and more than the efficient output is produced.

2. (Rosen 9.7) To work this problem, you'll need a calculator than can take logarithms or a spreadsheet program. Suppose that your utility function is $U=\ln(4I)$, where I is the amount of income you make in a given year. Suppose that you typically make \$30,000 per year, but there is a 5 percent chance that, in the next year, you will get sick and lose \$20,000 in income due to medical costs.

- a. What is your expected utility if you do not have insurance to protect against this adverse event?
- b. Suppose you can buy insurance that will cover your losses if you get sick. What would be the actuarially fair premium? What is your expected utility if you buy the insurance policy?
- c. What is the most that you'd be willing to pay for this policy?

ANS

a. There is a 95 percent chance of no illness, in which case income is \$30,000, and a 5 percent chance of illness, in which case income is \$10,000 because of the \$20,000 loss. Thus, expected income is $(0.95)(30,000) + (0.05)(10,000) = 29,000$. The utility of having income of \$29,000 with certainty is 11.66, but the expected utility is only $(0.95)U(30,000) + (0.05)U(10,000) = (0.95)(11.695) + (0.05)(10.597) = 11.64$.

b. An actuarially fair premium would be \$1,000 since there is a one in twenty chance that the insurance company will have to cover losses of \$20,000. If the individual buys insurance for \$1,000, then they have certain income of \$29,000 and the utility of \$29,000 is 11.66.

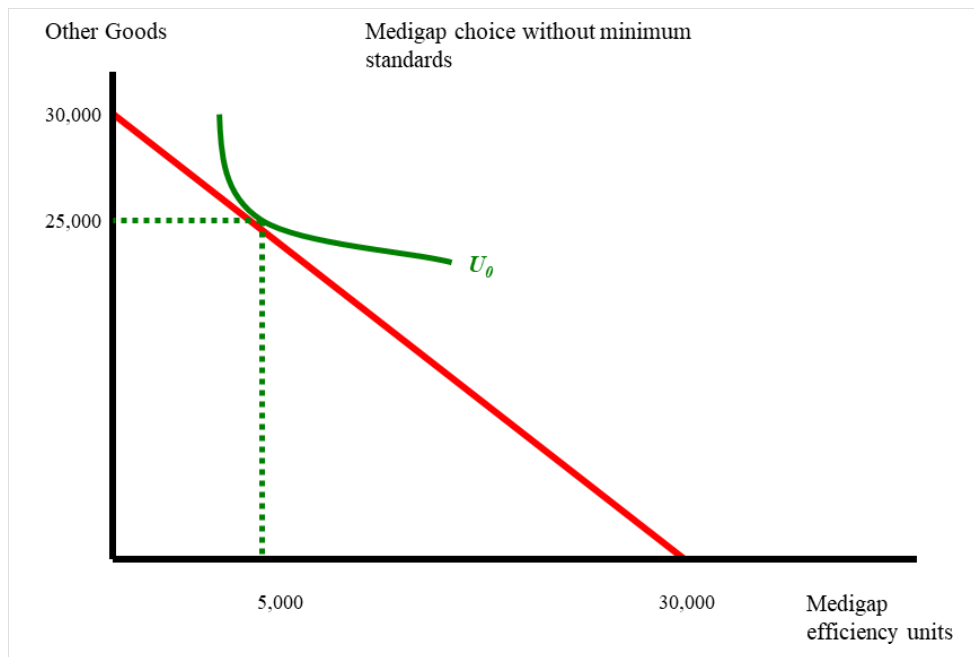
c. Setting the expected utility equal to 11.64 and solving for income yields approximately \$28,388, indicating that the individual is indifferent between bearing the risk and having expected income of \$29,000 or purchasing insurance with a certain income of \$28,388. If the insurance costs $\$30,000 - \$28,388 = \$1,612$, the individual is indifferent between having insurance and not having insurance, so \$1,612 is the most they'd be willing to pay (allow some difference for rounding).

3. (Rosen 10.5) Medicare recipients can purchase supplemental private insurance (known as Medigap insurance) to fill the gap in coverage left by Medicare. This gap includes copayments, deductibles, and prescription drug expenses not covered by Medicare. Several years ago, the government enacted regulations that specify minimum standards for items that Medigap policies must cover. This made the policies more expensive, and as a consequence, about 25 percent of the elderly who would have purchased more Medigap insurance purchased none at all [Finkelstein, 2004].

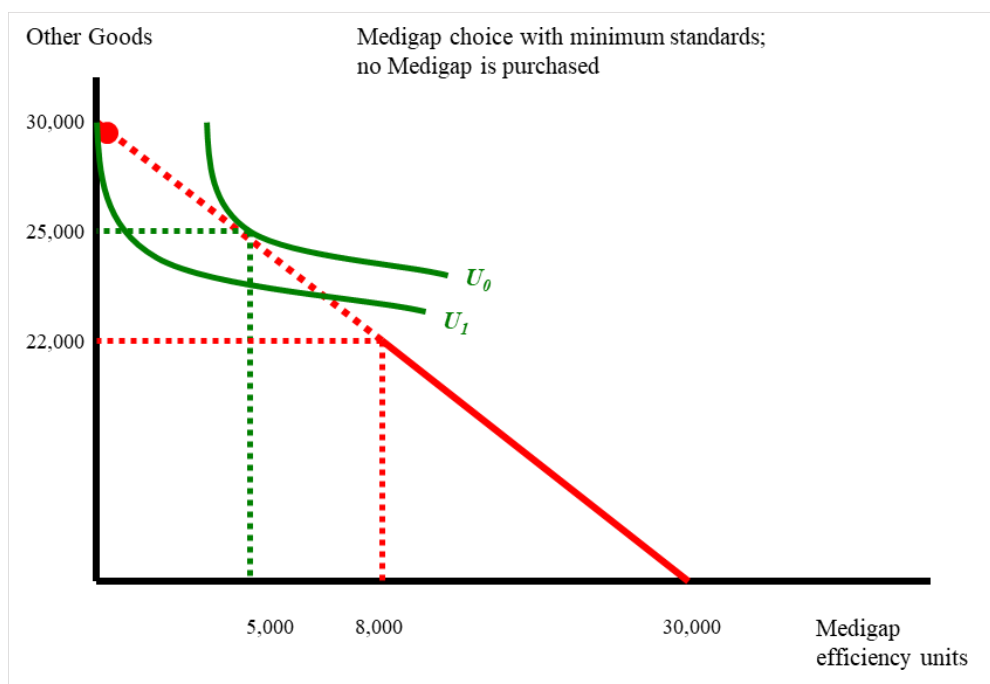
Consider an individual who consumes two goods: “insurance” and “all other goods.” The cost of a unit of Medigap insurance is \$1, as is the cost of a unit of all other goods. Sketch a budget constraint and set of indifference curves that are consistent with the following scenario: In an unregulated market, an individual with a \$30,000 income purchase \$5,000 worth of Medigap insurance. The government then puts mandates on Medigap policies that raise their minimum price to \$8,000; that is, the individual must purchase at least \$8,000 units of Medigap insurance or none at all. After considering the matter, the individual decides to go without Medigap insurance.

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The budget constraint initially has units of Medigap on the x-axis, and other goods on the y-axis. Given initial prices of \$1 per unit for each good, and \$30,000 of income, the budget constraint has a slope of -1, and the intercepts on both axes are at 30,000 units. It is assumed that the initial utility maximizing bundle consumes 5,000 units of Medigap, hence the indifference curve is tangent at (5000, 25000). All of this is illustrated in the figure below.



After the “minimum Medigap” mandate, the consumer can either choose 0 units of Medigap or 8,000 or more units of Medigap. Thus, part of the budget constraint is eliminated (though the overall shape remains the same as before). After the mandate, the point (0, 30000) is available, as well as all of the points to the southeast of the point (8000, 22000). Clearly, the person’s utility must fall since the preferred choice, (5000, 25000) is no longer available. If the person attains a higher level of utility as (0, 30000) compared with (8000, 22000), the person chooses to not purchase Medigap. In this case, the marginal rate of substitution is no longer equal to the price ratio. This is illustrated below.



4. 請說明下列現行全民健保現行部分負擔制度

- a. 門診 vs. 住院
- b. 低收入戶及醫療資源貧瘠地區

ANS

何謂部分負擔制度？

https://www.nhi.gov.tw/News_Content.aspx?n=EFF40B3ED686AC1A&sms=507DCB7041364C26&s=65F44762F60D2FA7

a. 門診部分負擔及住院部分負擔

https://www.nhi.gov.tw/News_Content.aspx?n=EFF40B3ED686AC1A&sms=507DCB7041364C26&s=3E8E4F90CE38C77B

b. (請參照「(二)免除部分負擔的對象」)

https://www.nhi.gov.tw/News_Content.aspx?n=EFF40B3ED686AC1A&sms=507DCB7041364C26&s=3E8E4F90CE38C77B

5. 請以 200 字介紹全民健保醫療費用總額支付制度

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https://www.nhi.gov.tw/Content_List.aspx?n=3CEE35FD85C63A93&topn=5FE8C9FEAE863B46